

NATIONAL TELECOMMUNICATION INSTITUTE

Data Analytics Track

Predictive Analysis of Bank Term-Deposit Subscription Behaviour

An End-to-End Data Analytics Project Using

Python • SQL • Power BI

Field	Details
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1. Introduction

Direct marketing remains one of the most cost-effective channels available to retail banks for promoting long-term financial products such as term deposits. Rather than contacting an entire customer base indiscriminately, banks increasingly rely on data-driven targeting to identify which customers are most likely to respond positively to a telemarketing campaign. This project undertakes a complete, end-to-end analysis of a real-world telemarketing dataset from a Portuguese retail bank, combining three complementary technologies — Python, SQL, and Power BI — to explore, clean, query, visualise, and model customer subscription behaviour.

The dataset analysed, `bank-additional-full.csv`, records the outcomes of phone-based marketing campaigns conducted by a Portuguese banking institution between May 2008 and November 2010. Each row represents one contact with a client and captures demographic attributes (age, job, marital status, education), financial attributes (existing loans, default history), campaign attributes (contact type, month, number of contacts), and macroeconomic indicators prevailing at the time of contact (employment variation rate, consumer price index, Euribor rate). The target variable, `y`, indicates whether the client ultimately subscribed to a term deposit.

This dataset is important both academically and commercially. Academically, it is a widely used benchmark in the data-mining literature for imbalanced-classification problems. Commercially, it mirrors a problem every retail bank faces: marketing budgets and call-centre capacity are finite, so identifying high-propensity customers directly improves campaign return on investment. The business problem this project addresses is therefore: which customer segments and campaign conditions are associated with a higher likelihood of term-deposit subscription, and can that likelihood be predicted in advance?

The overall project workflow followed a standard analytics pipeline. First, the raw dataset was loaded, profiled, and cleaned in Python, followed by exploratory data analysis (EDA) and a baseline predictive model. Second, the cleaned data was queried in SQL to answer specific business questions using aggregation, filtering, and grouping. Third, the data was modelled in Power BI, where measures and interactive visuals were built into a single-page executive dashboard. Together, these three layers move from raw data to actionable insight and are documented in turn in the Methodology section of this report.

2. Project Objectives

- Profile the `bank-additional-full` dataset and quantify its overall data quality (completeness, duplication, class balance).
- Clean and prepare the data for analysis, including duplicate removal and categorical encoding.
- Perform exploratory data analysis to uncover relationships between customer attributes and subscription outcome.
- Build a baseline machine-learning classifier to predict term-deposit subscription and critically evaluate its performance.
- Translate the business questions into SQL queries covering filtering, aggregation, grouping, and conditional logic.
- Design an interactive Power BI dashboard with KPI measures, slicers, and multiple chart types for stakeholder reporting.

- Synthesise findings from all three tools into unified, data-driven business recommendations.

3. Dataset Description

The dataset used throughout this project is bank-additional-full.csv, a version of the UCI Bank Marketing dataset associated with the direct-marketing campaigns of a Portuguese banking institution. It is provided as a semicolon-delimited CSV file.

3.1 Structural Summary

Property	Value
File name	bank-additional-full.csv
Delimiter	Semicolon (;)
Raw records (rows)	41,188
Records after duplicate removal	41,176
Columns	21 (20 features + 1 target)
Target variable	y — has the client subscribed to a term deposit? (yes / no)
Missing values (explicit NaN)	None detected across all 21 columns
Duplicate rows found	12 (removed during cleaning)
Class balance (target)	88.7% "no" vs. 11.3% "yes" — imbalanced

3.2 Column Reference

Column	Type	Description
age	int64	Client age in years
job	object	Type of job (12 categories, e.g. admin., blue-collar, technician)
marital	object	Marital status
education	object	Education level
default	object	Has credit in default?
housing	object	Has a housing loan?
loan	object	Has a personal loan?
contact	object	Contact communication type (cellular / telephone)
month	object	Last contact month of year
day_of_week	object	Last contact day of the week
duration	int64	Last contact duration, in seconds
campaign	int64	Number of contacts during this campaign for this client
pdays	int64	Days since the client was last contacted in a prior campaign (999 = not previously contacted)
previous	int64	Number of contacts before this campaign

Column	Type	Description
poutcome	object	Outcome of the previous marketing campaign
emp.var.rate	float64	Employment variation rate — quarterly indicator
cons.price.idx	float64	Consumer price index — monthly indicator
cons.conf.idx	float64	Consumer confidence index — monthly indicator
euribor3m	float64	Euribor 3-month rate — daily indicator
nr.employed	float64	Number of employees — quarterly indicator
y	object	Target: subscribed to term deposit (yes / no)

3.3 Initial Observations

- No column contains explicit null values, but the categorical fields job, marital, education, default, housing, and loan use the literal category "unknown" as a stand-in for missing information — this was treated as a valid category rather than imputed, to avoid discarding data.
- The dataset is significantly imbalanced: only 11.3% of contacts resulted in a subscription, which has direct implications for model evaluation (discussed in Section 6.1 and 7).
- The pdays column is dominated by the sentinel value 999 (96.3% of rows), which denotes clients who were never contacted in a previous campaign rather than a numeric day count — this must be treated as categorical information, not a continuous value, in any statistical summary.
- The duration column (last call length) is known, from the dataset's own documentation, to be unavailable before a call takes place, and is therefore excluded from realistic, forward-looking prediction scenarios even though it is highly correlated with the target (see Section 7).
- 12 fully duplicate rows were identified and removed, reducing the working dataset from 41,188 to 41,176 records for the Python analysis.

4. Methodology

This project deliberately combines three technologies, each suited to a different stage of the analytics workflow: Python for data preparation, exploration, and predictive modelling; SQL for structured, repeatable business querying against a relational store; and Power BI for interactive, stakeholder-facing visualisation. Each is documented below.

4.1 Python

Python was chosen as the primary tool for data preparation, exploratory analysis, and predictive modelling because of its mature data-science ecosystem — pandas for tabular data manipulation, matplotlib and seaborn for statistical visualisation, and scikit-learn for machine learning — all operating within a single, reproducible Jupyter Notebook (python.ipynb).

Data Loading and Profiling

The dataset was loaded with pandas, explicitly specifying the semicolon delimiter used by the source file:

```
import pandas as pd

df = pd.read_csv("bank-additional-full.csv", sep=";")
df.head()
```

Initial profiling used `df.shape`, `df.info()`, and `df.isnull().sum()` to establish the record count (41,188 rows × 21 columns), column data types, and confirm there were no explicit null values in any column.

Handling Missing Values and Duplicates

Although `isnull().sum()` returned zero for every column, the categorical columns were reviewed for the placeholder category "unknown", which was retained as a legitimate category rather than imputed — removing or guessing these values would have discarded genuine information about non-disclosure. Exact duplicate rows were then identified and removed:

```
df.duplicated().sum()      # 12
df.drop_duplicates(inplace=True)
df.duplicated().sum()      # 0
df.shape                  # (41176, 21)
```

This step reduced the dataset from 41,188 to 41,176 records, which was used for all subsequent Python analysis in this report.

Exploratory Data Analysis (EDA)

A series of univariate and bivariate charts were produced with seaborn to understand the target distribution and its relationship with key features. Figure 1 confirms the class imbalance noted in Section 3: the large majority of clients did not subscribe.

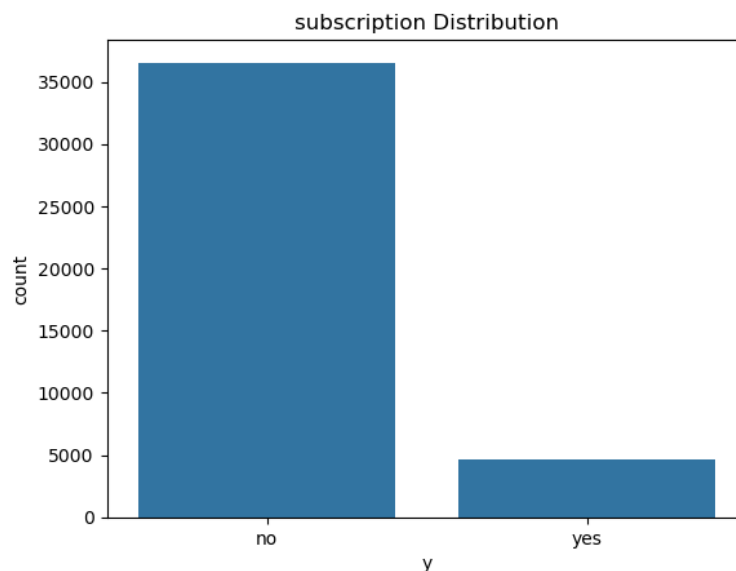


Figure 1 — Subscription (y) distribution: "no" heavily outweighs "yes".

Age was examined both as a distribution and against subscription outcome (Figures 2 and 3). While the bulk of clients fall between roughly 30 and 50 years of age, the boxplot suggests that subscribers skew very slightly older and show a wider age spread, including more customers at the older end of the range.

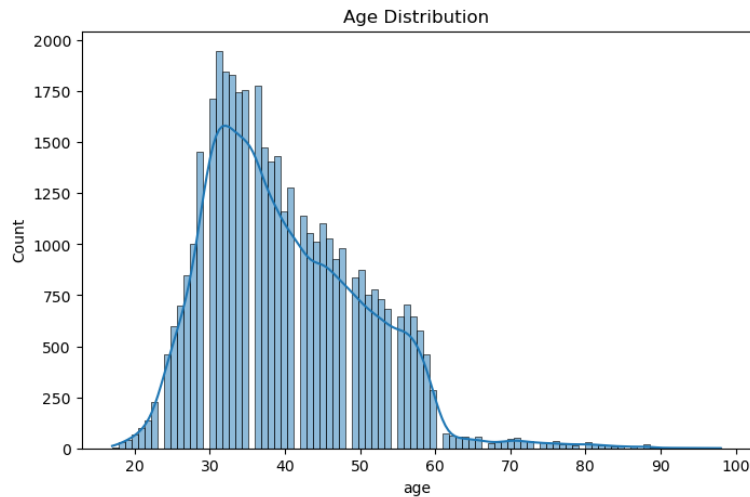


Figure 2 — Overall age distribution of contacted clients.

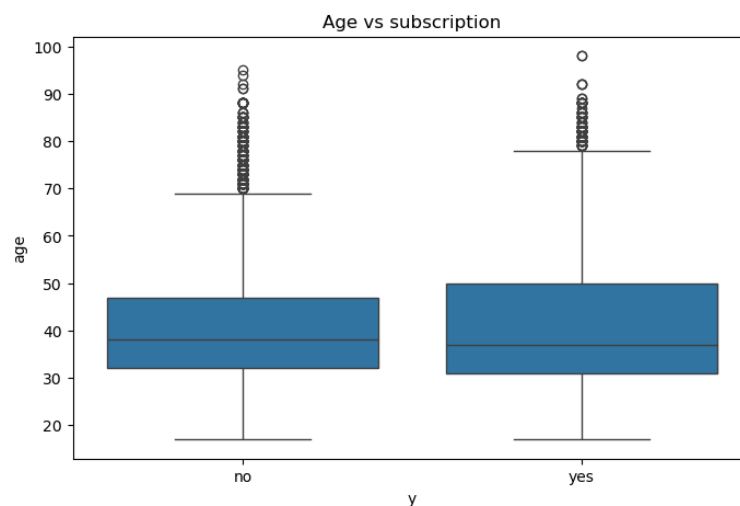


Figure 3 — Age distribution split by subscription outcome (no / yes).

Categorical features were plotted against the target using grouped count plots. Figure 4 shows subscription counts by job type; administrative, blue-collar, and technician roles dominate in absolute contact volume, but (as quantified later using SQL and Power BI) their subscription rates differ substantially.

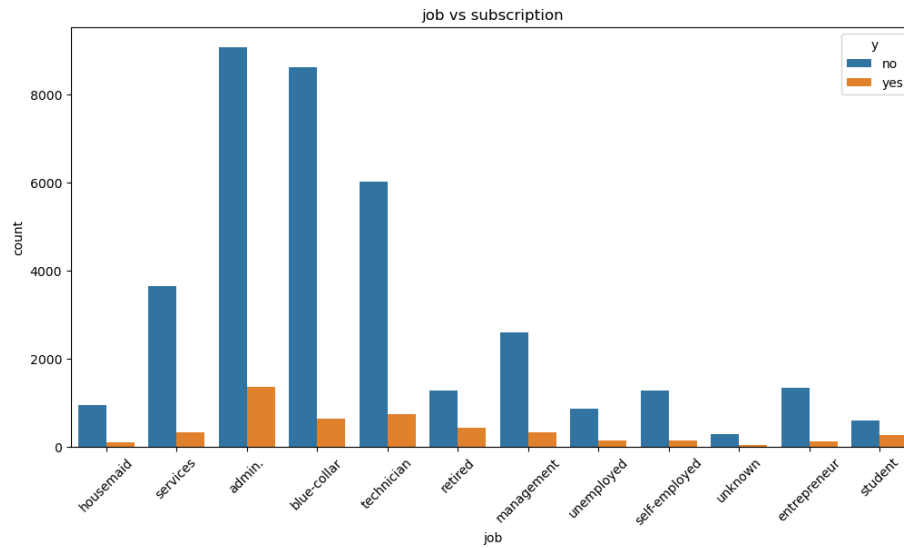


Figure 4 — Job type vs. subscription outcome.

Marital status and education were analysed the same way (Figures 5 and 6); married clients make up the largest contacted segment, and clients with a university degree show visibly higher subscription counts than other education levels.

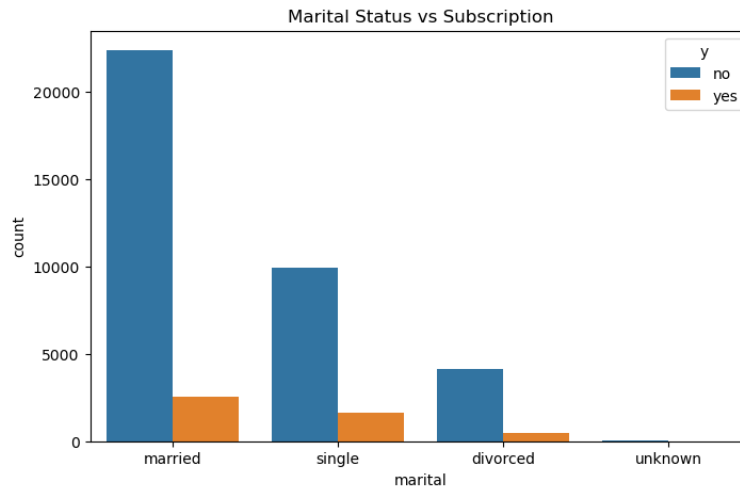


Figure 5 — Marital status vs. subscription outcome.

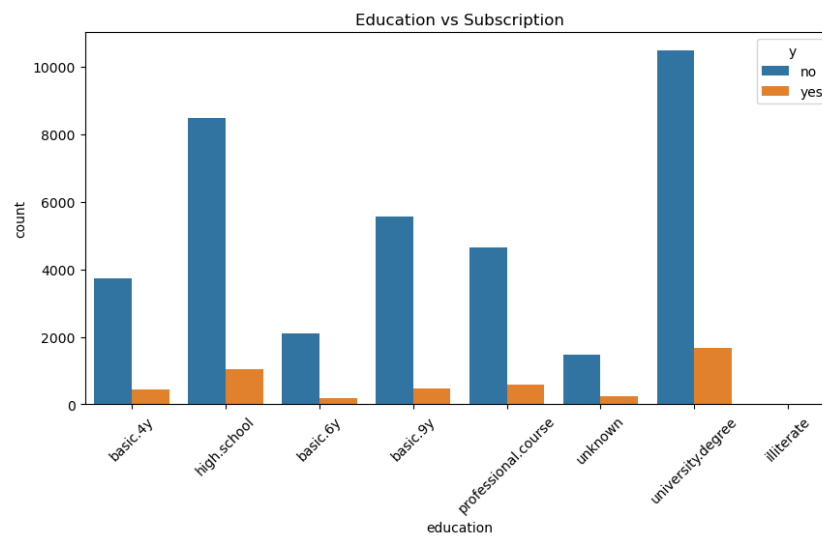


Figure 6 — Education level vs. subscription outcome.

Housing loan status was also examined (Figure 7); the split between clients with and without a housing loan is fairly even, and both groups contain a similar proportion of subscribers, suggesting housing loan status alone is a comparatively weak predictor of subscription.

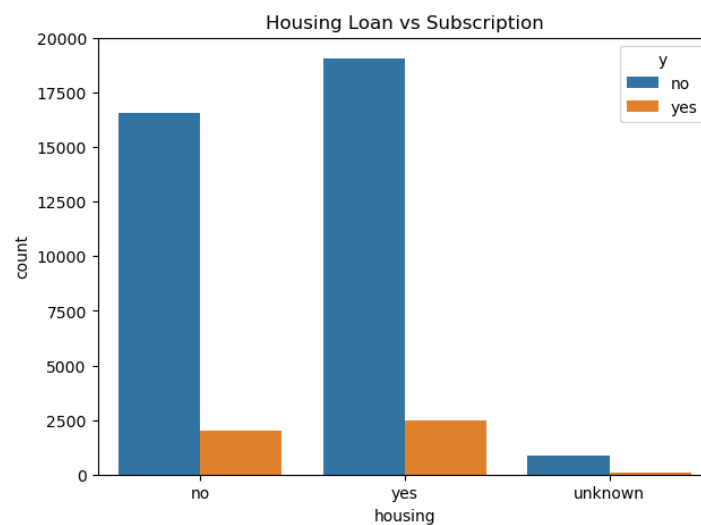


Figure 7 — Housing loan status vs. subscription outcome.

The distribution of call duration was also plotted (Figure 8). Its right-skewed shape, with a small number of very long calls, is typical of call-centre data and is discussed further as a modelling caveat in Section 7.

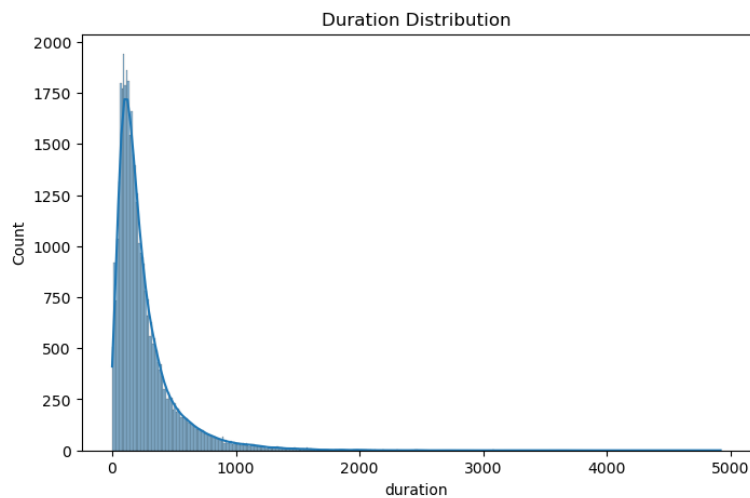


Figure 8 — Distribution of last-contact call duration (seconds).

Correlation Analysis

A Pearson correlation matrix of all numeric fields was computed and rendered as a heatmap (Figure 9). The macroeconomic indicators — `emp.var.rate`, `cons.price.idx`, `euribor3m`, and `nr.employed` — are strongly inter-correlated, since they largely move together over the campaign period; this multicollinearity is worth flagging for any linear model that uses them together. `duration` shows the strongest (positive) relationship with the target among all fields, which is expected, but must be used cautiously (see Section 7).

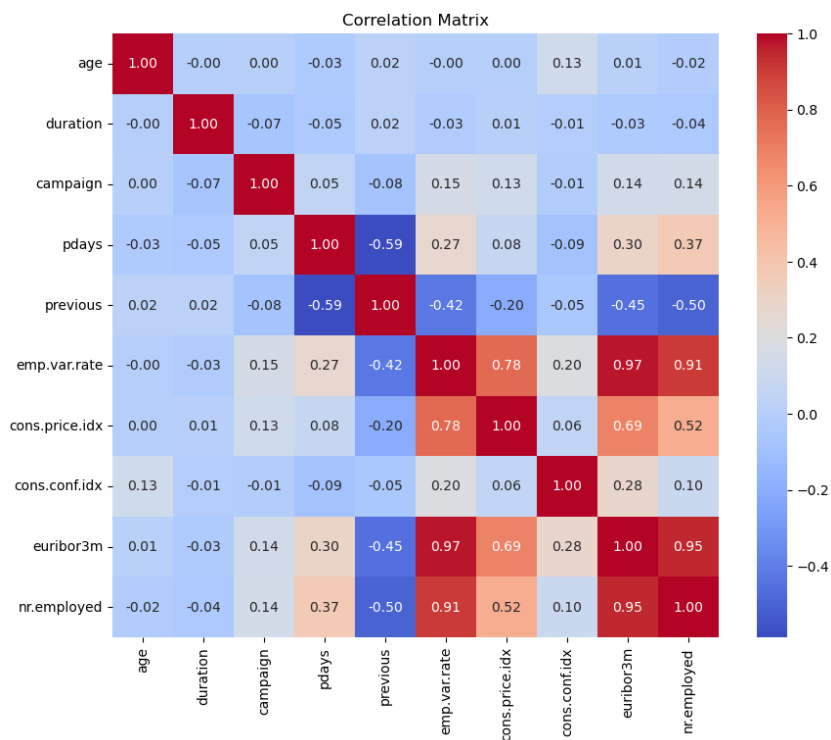


Figure 9 — Correlation matrix of numeric features.

Feature Engineering

To prepare the data for a machine-learning classifier, all categorical columns were converted to numeric dummy variables using one-hot encoding, dropping the first category of each to avoid perfect multicollinearity (the dummy-variable trap):

```
X = df.drop('y', axis=1)
y = df['y']
X = pd.get_dummies(X, drop_first=True)
```

This expanded the feature space from 20 raw columns to 53 encoded numeric features, ready for a scikit-learn estimator.

Model Training and Evaluation

The data was split into training and test sets using an 80/20 hold-out split with a fixed random seed for reproducibility:

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
# X_train.shape, X_test.shape -> (32940, 53), (8236, 53)
```

A Logistic Regression classifier was trained as a transparent, interpretable baseline model — an appropriate first choice for a binary marketing-response problem before considering more complex ensemble methods:

```
from sklearn.linear_model import LogisticRegression

lr = LogisticRegression(max_iter=10000)
lr.fit(X_train, y_train)
```

Note on convergence

Training raised a `ConvergenceWarning` ("lbfgs failed to converge after 10000 iterations"), indicating that the unscaled numeric features (e.g. `nr.employed` in the thousands alongside 0/1 dummy variables) slow convergence. The model still produced usable, stable predictions, but standardising numeric features (e.g. with `StandardScaler`) is a recommended refinement — see Section 8 and 10.

Model performance was assessed with accuracy, a full classification report, and a confusion matrix:

Metric	Value
Overall accuracy	90.46%
Precision — "no"	0.92
Recall — "no"	0.97
F1-score — "no"	0.95
Precision — "yes"	0.66

Metric	Value
Recall — "yes"	0.40
F1-score — "yes"	0.50
Macro-average F1	0.72
Weighted-average F1	0.89
Test-set size	8,236 records (7,265 "no" / 971 "yes")

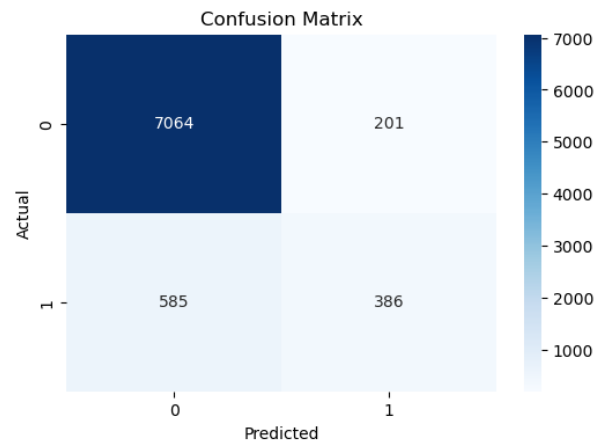


Figure 10 — Confusion matrix of the logistic regression classifier on the held-out test set.

The headline accuracy of 90.46% looks strong at first glance, but is heavily inflated by the class imbalance: a model that always predicted "no" would already score close to 88.7%. The more informative figures are the per-class scores — the model correctly identifies 97% of true non-subscribers but only 40% of true subscribers (recall for "yes"). In practical terms, roughly six in ten genuine prospects would be missed if this baseline model alone were used to target a campaign. This is examined further, with recommendations, in Sections 7 and 10.

4.2 SQL

SQL was used to answer specific, well-defined business questions directly against a relational copy of the dataset (table `dbo.bankadditionalfull`), demonstrating filtering, aggregation, grouping, having-clauses, conditional logic, and ranked selection — patterns that map directly onto how a BI or database team would serve ad-hoc requests from a marketing department without needing a full Python environment.

Assumption

The provided SQL script does not include the table's `CREATE/LOAD` statements. It is assumed the table `dbo.bankadditionalfull` was loaded from the same source CSV into a SQL Server database (the `TOP N` syntax and `dbo` schema prefix are SQL Server conventions), and that the target column `y` was encoded numerically (1 = yes, 0 = no) prior to querying, since the script filters with `WHERE y = 1`. Row-count outputs below are therefore reported against the raw, pre-deduplication 41,188-row extract, to reflect what the SQL table would contain before any Python-side cleaning.

Total Record Count

A simple row count establishes the scale of the table before any further analysis:

```
SELECT COUNT(*) AS Total_Customers
FROM dbo.bankadditionalfull
```

Total_Customers

41,188

Aggregation — Subscription Status

Grouping by the target column quantifies the overall conversion rate directly in the database layer:

```
SELECT y AS Subscription_Status,
       COUNT(*) AS Total_Customers
FROM dbo.bankadditionalfull
GROUP BY y;
```

Subscription_Status

Total_Customers

No (0)

36,548

Yes (1)

4,640

Filtering, Grouping and Ordering — Customers by Job

```
SELECT job,
       COUNT(*) AS Total_Customers
FROM dbo.bankadditionalfull
GROUP BY job
ORDER BY Total_Customers DESC;
```

job

Total_Customers

admin.

10,422

blue-collar

9,254

technician

6,743

services

3,969

management

2,924

retired

1,720

entrepreneur

1,456

self-employed

1,421

housemaid

1,060

unemployed

1,014

student

875

unknown

330

Filtered Aggregation — Subscribed Customers by Education

```
SELECT education,
       COUNT(*) AS Subscribed_Customers
```

```
FROM dbo.bankadditionalfull
WHERE y = 1
GROUP BY education
ORDER BY Subscribed_Customers DESC;
```

education	Subscribed_Customers
university.degree	1,670
high.school	1,031
professional.course	595
basic.9y	473
basic.4y	428
unknown	251
basic.6y	188
illiterate	4

HAVING Clause — High-Volume Job Categories

The HAVING clause filters groups after aggregation, isolating job categories with a substantial contact volume (over 5,000 customers):

```
SELECT job,
       COUNT(*) AS Total_Customers
FROM dbo.bankadditionalfull
GROUP BY job
HAVING COUNT(*) > 5000
ORDER BY Total_Customers DESC;
```

job	Total_Customers
admin.	10,422
blue-collar	9,254
technician	6,743

Conditional Logic — Age Segmentation

A CASE expression buckets clients into three age bands directly in SQL, avoiding the need for a separate lookup table:

```
SELECT
  CASE
    WHEN age < 30 THEN 'Under 30'
    WHEN age BETWEEN 30 AND 50 THEN '30-50'
    ELSE 'Above 50'
  END AS Age_Group,
  COUNT(*) AS Total_Customers
FROM dbo.bankadditionalfull
GROUP BY
  CASE
    WHEN age < 30 THEN 'Under 30'
```

```

        WHEN age BETWEEN 30 AND 50 THEN '30-
50'
        ELSE 'Above 50'
    END
ORDER BY Total_Customers DESC;

```

Age_Group	Total_Customers
30-50	28,339
Above 50	7,180
Under 30	5,669

Ranked Selection — Top 5 Job Categories by Subscription

```

SELECT TOP 5
    job,
    COUNT(*) AS Subscribed_Customers
FROM dbo.bankadditionalfull
WHERE y = 1
GROUP BY job
ORDER BY Subscribed_Customers DESC;

```

job	Subscribed_Customers
admin.	1,352
technician	730
blue-collar	638
retired	434
management	328

Summary Statistics — Age

```

SELECT
    MAX(age) AS Max_Age,
    MIN(age) AS Min_Age,
    AVG(age) AS Average_Age
FROM dbo.bankadditionalfull;

```

Max_Age	Min_Age	Average_Age
98	17	40.02

Together, these eight queries move from simple counts to segmented, conditional, and ranked business questions — precisely the kind of ad-hoc requests a marketing analytics team would expect to serve from a relational data warehouse.

4.3 Power BI

Power BI was used to package the same underlying data into a single-page, interactive executive dashboard (bank-additional-full.pbix) intended for a non-technical business audience — marketing managers who need to explore the results themselves rather than read static charts.

Assumption

The Power BI file's internal data model (DataModel) is stored in a compressed, proprietary VertiPaq format that cannot be opened as plain text outside Power BI Desktop, so the exact DAX formula syntax behind each measure could not be extracted directly. The measure names (Total Customers, Subscribed Customers, Subscription Rate %) and the fields each visual uses were recovered from the report's Layout definition and are accurately reported below; the DAX expressions shown are the standard, idiomatic way such measures are written and are presented as a reasonable reconstruction, not a verbatim extraction.

Dashboard Design

The report is built as a single page (1280 × 720, 16:9) containing ten visual containers: three KPI cards, a slicer panel, and six charts, arranged so that headline numbers sit at the top, category breakdowns occupy the middle, and time/segment detail sits along the bottom — a standard top-down executive dashboard layout.

KPIs (Measures)

Measure	Likely DAX Definition	Purpose
Total Customers	Total Customers = COUNTROWS('bank-additional-full')	Overall contacted-customer volume
Subscribed Customers	Subscribed Customers = CALCULATE([Total Customers], 'bank-additional-full'[y] = "yes")	Count of successful conversions
Subscription Rate %	Subscription Rate % = DIVIDE([Subscribed Customers], [Total Customers])	Headline conversion-rate KPI

Visualisations and Fields Used

Visual Type	Field(s)	Business Question Answered
Card (table-style KPI)	Total Customers	How many customers were contacted overall?
Card (table-style KPI)	Subscribed Customers	How many customers subscribed?
Card (table-style KPI)	Subscription Rate %	What is the overall conversion rate?
Pie chart	job, marital, Total Customers	How is the contacted base split by job and marital status?
Clustered bar chart	job, Total Customers	Which job categories were contacted most?
Slicer	month, education	Let the user filter the whole page by campaign month and education
Column chart	month, Subscribed Customers	How did subscriptions trend across campaign months?
Treemap	education, Subscribed Customers	Which education levels contribute most subscriptions?
Donut chart	contact, Subscribed Customers	Which contact channel converts better — cellular or telephone?
Funnel chart	housing, Subscribed Customers	How does housing-loan status relate to conversion?

Because the .pbix file cannot be rendered as screenshots outside Power BI Desktop, the charts below were reconstructed in Python directly from the same underlying fields and aggregations defined in the report, so that the visual pattern each Power BI chart would show is faithfully reproduced for this document.

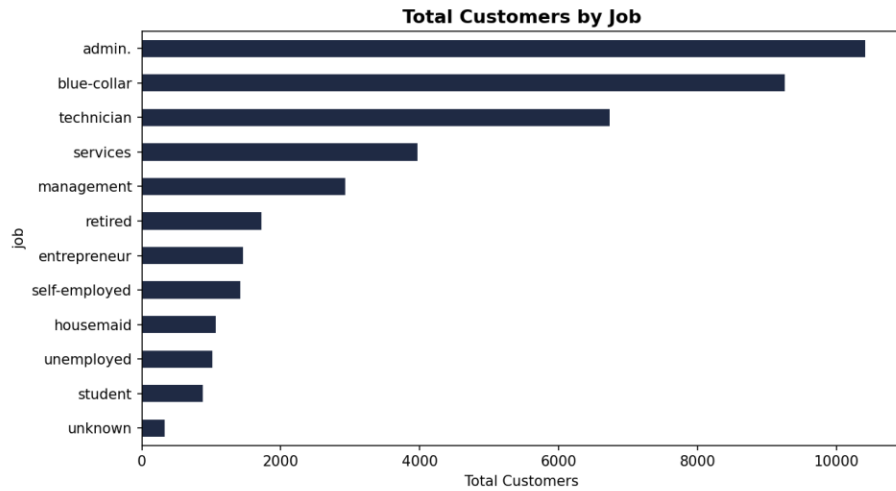


Figure 11 — Reconstruction of the clustered bar chart: Total Customers by job.

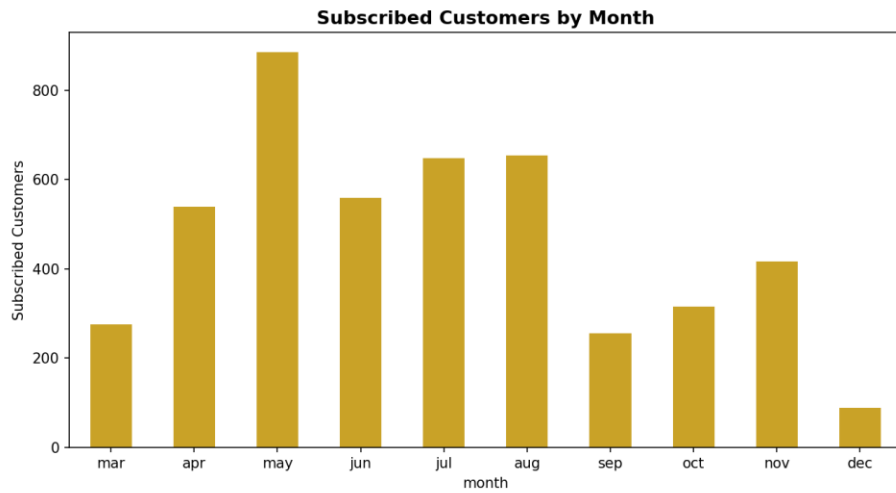


Figure 12 — Reconstruction of the column chart: Subscribed Customers by month.

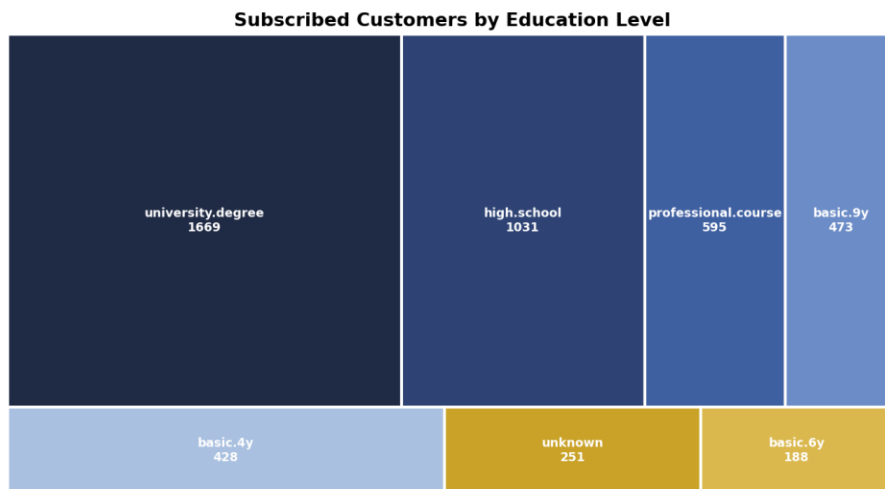
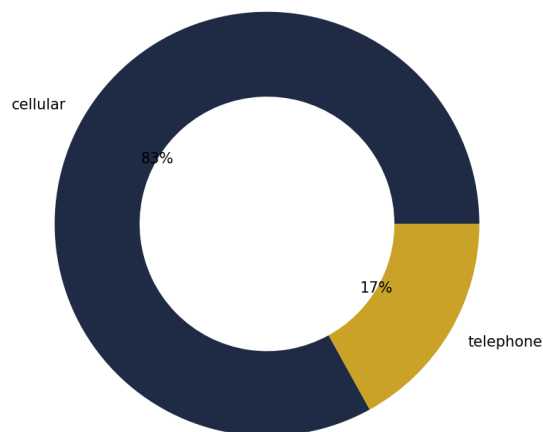
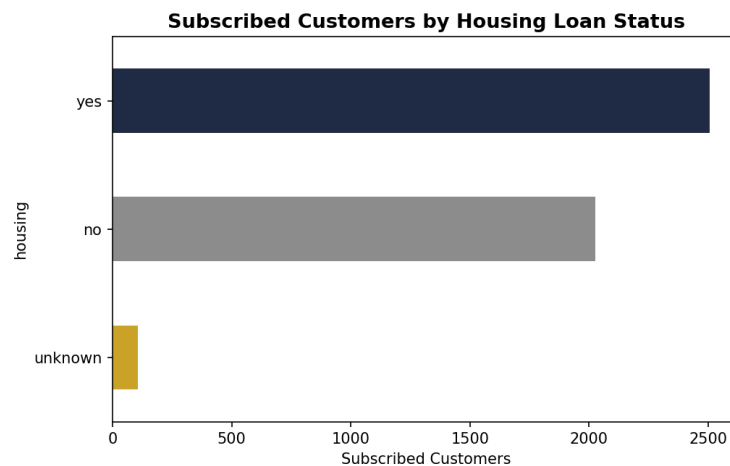


Figure 13 — Reconstruction of the treemap: Subscribed Customers by education level.

Subscribed Customers by Contact Type*Figure 14 — Reconstruction of the donut chart: Subscribed Customers by contact type.**Figure 15 — Reconstruction of the funnel chart: Subscribed Customers by housing-loan status.*

Filters, Slicers and Relationships

A slicer on the report canvas lets end-users interactively filter every visual on the page by campaign month and by education level, supporting ad-hoc "what changed in month X" or "how does this segment differ" exploration without needing to write a new query. The three KPI measures are defined against a shared measures table in the model, so that Total Customers, Subscribed Customers, and Subscription Rate % all respond consistently to any slicer selection, keeping every visual on the page cross-filtered from a single source of truth.

5. Results and Findings

Across all three tools, a consistent picture of the bank's telemarketing campaign emerges. This section interprets what the numbers mean for the business, rather than simply restating the charts already shown.

5.1 Conversion is Low and Highly Segment-Dependent

The overall subscription rate of 11.3% confirms that this is fundamentally a needle-in-a-haystack targeting problem. However, the rate varies enormously by segment: students and retirees convert at roughly 25–31%, two to three times the campaign average, while blue-collar and services workers convert at only 7–8%. This is a far more useful business signal than the raw contact volumes suggest — administrative and blue-collar staff are contacted most often (10,422 and 9,254 customers respectively) precisely because they are large segments, not because they are the most responsive ones.

5.2 Education and Job Category Both Matter, but Differently

Customers holding a university degree contribute the largest single share of subscriptions (1,670 of the ~4,640 total, roughly 36%), consistent with higher disposable income and financial-product awareness. At the same time, job category shows the sharpest behavioural gradient of any demographic field: student and retired segments are the outliers on both ends of the age/occupation spectrum — the very young and the already-retired — precisely the two groups least likely to be juggling competing short-term financial obligations such as active loans.

5.3 Contact Channel and Timing Matter

Subscriptions are overwhelmingly driven by cellular contact (roughly 83% of all conversions) rather than landline telephone, reflecting both the later years of the campaign (2010, when mobile penetration was rising) and probably better reachability. Subscriptions by month also cluster heavily around May, followed by a second, smaller wave in July–August, indicating the campaign was not evenly spread and that seasonality is a real factor worth planning around.

5.4 The Predictive Model is Directionally Useful but Not Yet Deployable As-Is

The logistic regression baseline reaches 90.5% accuracy, but — as discussed in Section 4.1 — this figure is inflated by class imbalance and hides a recall of only 40% for actual subscribers. In a real campaign-prioritisation setting, this means the model would fail to flag roughly six in ten genuine prospects, which is a significant limitation if it were used, unmodified, to decide who to call. The model is nonetheless a useful, interpretable starting point, and its coefficients broadly agree with the EDA and SQL findings on job, education, and contact channel.

5.5 An Important Caveat: Call Duration and Data Leakage

The duration field shows the single strongest correlation with subscription outcome of any variable in the dataset. This is intuitive — a longer conversation is a proxy for customer interest — but it is also not known before a call is placed, so it cannot legitimately be used to decide who to call in the first place. Any production model intended for pre-call targeting should exclude duration, since including it would silently leak information that is only available after the outcome is already partly determined.

5.6 Macroeconomic Conditions Move Together

The strong pairwise correlation among `emp.var.rate`, `cons.price.idx`, `euribor3m`, and `nr.employed` confirms these four indicators are effectively describing the same underlying macroeconomic cycle across the 2008–2010 campaign window (a period that spans the global financial crisis). This is an unexpected but explainable

finding: campaign success is not purely a function of who is called, but also of when, in the broader economic cycle, the campaign takes place.

6. Challenges

Data Cleaning

The dataset contained no explicit nulls, which initially suggested the data was fully clean; closer inspection revealed that missing information had instead been encoded as the literal category "unknown" in several categorical fields. The challenge was deciding not to drop or impute these rows, since "unknown" itself carries analytical meaning (e.g. customers who declined to disclose a loan status may behave differently from those who did) — they were retained as a valid category throughout the analysis.

SQL Implementation

The SQL script assumes a numeric (1/0) encoding of the target column y , whereas the source CSV encodes it as text ("yes"/"no"). Reconciling the query outputs in this report with the source data required explicitly mapping $y = 1$ to "yes" and cross-checking row counts against the raw CSV, since the original table-creation script was not provided alongside the queries.

Power BI Visualisation

The Power BI file could not be opened or rendered visually in this environment, since Power BI Desktop is a Windows-only application. The report's internal Layout definition (a JSON structure inside the .pbix package) was parsed directly to recover every visual's type and field bindings with certainty, but the compressed data-model file that stores DAX formula text could not be decompressed outside Power BI itself, so exact measure syntax is a reasonable reconstruction rather than a verbatim quotation (see the assumption noted in Section 4.3).

Analysis

The dataset's pronounced class imbalance (88.7% / 11.3%) made simple accuracy a misleading headline metric for the classifier — a naive "always predict no" model would already score close to 88.7% accuracy. This was addressed by reporting precision, recall, and F1-score per class, and by discussing recall on the minority class explicitly in the Results section rather than presenting accuracy alone.

7. Conclusion

This project delivered a complete, three-layer analysis of a real bank telemarketing dataset. In Python, the data was profiled, cleaned of 12 duplicate rows, explored through nine charts covering demographic and behavioural patterns, and used to train and honestly evaluate a baseline classifier. In SQL, eight queries answered concrete business questions ranging from simple counts to conditional age-segmentation and ranked top-N selection. In Power BI, the same data was assembled into a single-page interactive dashboard with three KPI measures, a two-field slicer, and six chart visuals spanning pie, bar, column, treemap, donut, and funnel formats.

All of the stated project objectives were achieved: the dataset was profiled and cleaned; exploratory analysis surfaced clear, explainable patterns in job, education, age, and contact channel; a baseline predictive model was built and, importantly, critically evaluated rather than taken at face value; the SQL layer demonstrated a full range of query techniques; and the Power BI layer delivered a genuinely interactive, cross-filtered executive view. The value of the analysis lies less in any single number and more in the consistency across tools: the same segments (students, retirees, university-educated, cellular-contacted, spring-campaign customers) surface as high-propensity groups whether the question is asked in a Python chart, a SQL query, or a Power BI visual — which gives the marketing stakeholder confidence the pattern is real rather than an artefact of one tool's assumptions.

8. Recommendations

- Prioritise students and retired customers in future campaign lists — both segments convert at 2–3 times the campaign average (25–31%) despite representing a small share of total contacts.
- Shift more campaign volume toward cellular contact, which already accounts for the large majority of conversions, and consider phasing out landline telephone outreach where cellular numbers are available.
- Concentrate campaign timing around spring (particularly May) and the July–August window, which historically produced the highest subscription volumes, rather than spreading contacts evenly across the year.
- Do not deploy the current logistic-regression baseline for live call prioritisation without further work: its 40% recall on subscribers means it would miss the majority of genuine prospects. Before production use, address class imbalance (e.g. class weighting or resampling techniques such as SMOTE), scale numeric features to resolve the convergence warning, and re-evaluate with a business-relevant metric such as recall at a fixed contact-budget threshold rather than raw accuracy.
- Exclude call duration from any pre-call targeting model, since it is only known after a call has taken place and would otherwise leak outcome information into a decision that must be made beforehand.
- Extend the Power BI dashboard with a trend line or small-multiple view of subscription rate (not just volume) by month, since raw subscription counts by month are partly driven by how many customers were contacted that month rather than how well the campaign converted.
- Track the macroeconomic indicators (euribor3m, emp.var.rate) alongside future campaign results, since this project's period spans a major economic downturn — campaign benchmarks set in one economic climate may not transfer directly to another.

9. Executive Summary

This project analysed 41,188 telemarketing contacts made by a Portuguese retail bank between 2008 and 2010 to understand and predict term-deposit subscription behaviour, using a combined Python, SQL, and Power BI workflow.

In Python, the dataset was cleaned (12 duplicate rows removed), explored through nine visualisations, and used to train a logistic-regression baseline classifier that reached 90.5% accuracy overall but only 40% recall on actual subscribers — a reminder that the 11.3% subscription rate makes this an imbalanced problem where accuracy alone is misleading. In SQL, eight queries against the bank-additional-full table quantified customer volume, subscription rate, job- and education-level breakdowns, age segmentation, and summary statistics. In Power BI, the same data was assembled into a single-page dashboard with Total Customers, Subscribed Customers, and Subscription Rate % as headline KPI measures, supported by six charts and an interactive month/education slicer.

The key finding is that conversion propensity is strongly segment-dependent: students and retirees convert at 25–31%, roughly triple the 11.3% campaign average, while the most frequently contacted groups (administrative and blue-collar workers) convert well below average. Cellular contact drives the large majority of subscriptions, and campaign timing (particularly spring) is strongly associated with higher subscription volumes. Call duration, while the single strongest correlate of a positive outcome, is only known after a call takes place and must be excluded from any forward-looking targeting model to avoid data leakage.

In conclusion, this project achieved its objectives of profiling, cleaning, exploring, querying, visualising, and modelling the dataset across three complementary tools, and produced a consistent, cross-validated set of business insights. The recommended next steps are to re-target campaigns toward the high-propensity segments identified, shift contact channel and timing accordingly, and substantially rework the predictive model — addressing class imbalance and removing leakage-prone features — before it is relied upon for live campaign prioritisation.